

That is, the value of a cap less the value of a floor with the same strike is equal to the value of a swap in which we are paying fixed at a rate equal to the strike. Once again, it is assumed in this case that both the fixed side and the floating side of the swap are quoted using the same day count and frequency as the cap and floor. For example, if we are talking about caps and floors on 3-month LIBOR that pay quarterly money, then it is assumed that the swap pays quarterly money versus 3-month LIBOR quarterly money as well.

Cap-floor parity tells us that being long a cap and short a floor with the same strike is the same thing as paying fixed in a swap at a fixed rate equal to the strike. We can use the parity relationship in Equation 4.5 to express the relationship between caps, floors, and swaps in other equivalent ways. For example, “multiplying Equation 4.5 by -1 ” tells us that being short a cap and long a floor is the same as receiving in a swap.³

Consider now a cap and a floor on 3-month LIBOR with the same maturity and strike. Suppose the strike is such that the cap and the floor have the same value today. What is the significance of this strike? From cap-floor parity we know that being long a cap and short a floor on 3-month LIBOR is equivalent to paying fixed in a swap, where the fixed rate is quoted quarterly money. Since the cap and the floor have the same value, this means by cap-floor parity that the swap also has zero value. Thus, this particular strike must be the par quarterly money swap rate on a swap of the given maturity. To put it another way, it is the quarterly money fixed rate in the swap such that the swap is an $NPV = 0$ transaction, which is by definition the par swap rate.

4.1.2 Uses of Caps and Floors

The buyer of a cap or floor is protecting against an adverse move in the underlying rate. The buyer of a cap is protecting against rates moving higher, whereas a floor buyer is protecting against rates moving lower. It is important to remember that caps and floors are actually series of options, so the protection extends over multiple periods. Here are some traditional uses of caps and floors:

- A bank may have floating liabilities based on 3-month LIBOR. In order to protect itself from an increase in rates, the bank may purchase caps.⁴ The bank may partially offset the cost of the caps by selling lower strike

³Yet another perturbation of this relationship tells us that being long a cap and receiving fixed in a swap is the same thing as being long a floor.

⁴FAS (Financial Accounting Standards) 133, an accounting standard released in 1998, made it less attractive for banks to purchase caps, as doing so would potentially add to earnings volatility.

floors. Such a position that is long caps and short floors is known as a *collar*.

- A *mortgage servicer* is responsible for collecting monthly mortgage payments from borrowers, handling any escrow account associated with the loan, sending out late notices when necessary, and so on.⁵ That is, a mortgage servicer is responsible for administering the loan once it has been originated. In compensation for performing these duties, the mortgage servicer is typically given a piece of the monthly mortgage coupon, say, 25 bps, that is earned on the outstanding mortgage balance. As the mortgage balance decreases, the payments received by the servicer proportionately decrease. In the event of prepayment, the fee arising from that mortgage that is earned by the servicer completely disappears.

Servicing is a valuable asset. Most large originators of mortgages retain their mortgage servicing rights but hedge the asset distinctly from their other assets. Servicers face a tremendous amount of interest rate risk. If the market rallies after the servicing rights have been purchased, borrowers will have an increased incentive to prepay, and the outstanding balance on which the servicer earns his 25 bps may decline dramatically.⁶ In order to protect against this risk, servicers have sometimes purchased floors, which increase in value as the market rallies.⁷ Although servicers sometimes purchase floors on 3-month LIBOR, they have more traditionally purchased floors on either 10-year CMS or 10-year CMT. The prepayment of mortgages tends to be driven more by the 10-year part of the yield curve rather than on the front end, making floors on these indexes more appropriate hedges. Swaptions (which we will look at in detail in Chapter 5) are also used, as their liquidity allows for easier rebalancing and monetization.

- Consider a portfolio manager who purchases \$100mm of a 5-year LIBOR *capped floater*, a bond that pays quarterly coupons equal to 3-month LIBOR + 100 bps, *capped* at 9%. That is, each quarter LIBOR is set, and based upon this setting, the bond pays out a coupon equal to

$$\min(3\text{-month LIBOR} + 100 \text{ bps}, 9\%).$$

If the portfolio manager is concerned about an increase in rates, he may purchase a cap on 3-month LIBOR. Note that the *coupon cap* on

⁵For a quick review on mortgages, see Appendix D.

⁶For those who know a bit about mortgages, servicing rights are an asset that have similar economic characteristics to mortgage IO (interest only) strips.

⁷In recent years, many servicers have been less inclined to hedge with floors and more inclined to hedge with other instruments, including swaps.

this bond, the maximum coupon that is obtainable, is 9%. There is a short option position embedded in this bond that prevents the coupon from increasing beyond a certain coupon level. This option comes into the money when 3-month LIBOR is high. That is, the embedded option is a cap on 3-month LIBOR. The effective strike on the embedded LIBOR cap is $9\% - 1\% = 8\%$.⁸

The portfolio manager may instead decide to buy a *cap corridor* in which he buys one cap struck at 8% and simultaneously sells another struck at a higher strike, say, 10%. In this way, the portfolio manager would raise the effective coupon cap from 9% to 11% at a lower price than uncapping the bond completely.

4.1.3 An Embedded Cap Trade

Sometimes options are embedded into swaps (a concept that we explore in greater detail in Chapter 6), as the following example will demonstrate. Consider the trade shown in Figure 4.2, in which the dealer receives a fixed rate of 1.339% semi bond and pays an amount equal to the minimum of 3-month LIBOR and 3% quarterly Act/360 to the client for five years on a notional of \$100mm.

Let's take a closer look at what the dealer is paying in this trade. By paying the minimum of LIBOR and 3% instead of just paying LIBOR, the dealer is protected whenever LIBOR is above 3% on any reset date. That is, if LIBOR exceeds 3% on any reset date, the dealer will only have to pay 3%. As we will see, it is as if the dealer is paying LIBOR and has purchased a cap from the client struck at 3%.

⁸It is important to recognize that the strike on the LIBOR cap is 8%, and not 9%. First of all, we need to get one thing out of the way: the underlying on this embedded cap is 3-month LIBOR, not 3-month LIBOR + 100 bps. While one could certainly make the argument that the underlying includes the index of 100 bps, industry convention dictates the underlying is simply 3-month LIBOR. You wouldn't walk up to an options trader and ask for the price of a "3-month LIBOR + 100 bps cap." People would look at you like you had two heads.

With this established, we can figure out that the strike on the embedded LIBOR cap is 8% a couple of different ways. The first way is to just play around with some numbers: if LIBOR sets at 7.99%, the coupon will be 8.99%; if LIBOR sets at 8%, the coupon will be 9%; but if LIBOR sets at 8.01%, the coupon will still be 9%. Thus, the embedded option has a strike of 8%. Try the same thing starting with 8.99%, and you'll see the coupon stays at 9%, suggestive that the strike on the embedded option is not 9%.

Alternatively (and more formally), notice that

$$\min(\text{LIBOR} + 100 \text{ bps}, 9\%) = \text{LIBOR} + 100 \text{ bps} - \max(\text{LIBOR} - 8\%, 0\%).$$

Thus, we can equivalently look at the coupon on the bond as paying out 3-month LIBOR + 100 bps, less the payout of an 8% cap on 3-month LIBOR.



Counterparty pays:	1.339% semi on 30/360 basis
Client pays:	The minimum of 3mL and 3%, quarterly on Act/360 basis
Notional:	\$100mm
Maturity:	5 years

Current offer side	
5-year swap rate:	1.647% semi 30/360

Figure 4.2: Example of a Swap with an Embedded Cap

Notice, for example, that if LIBOR were to set at 3.01%, the dealer would normally have to pay out 3.01% if he were just paying LIBOR. But in this trade, the dealer will pay only $\min(3.01\%, 3\%) = 3\%$. We can view this equivalently as the dealer paying out the LIBOR rate of 3.01% and simultaneously getting the payoff from cap struck at 3%, which is $\max(3.01\% - 3\%, 0) = 1 \text{ bp}$. This means the net payout made by the dealer is $3.01\% - 1 \text{ bp} = 3\%$. We can formalize this way of thinking by noticing that

$$\min(\text{LIBOR}, 3\%) = \text{LIBOR} - \max(\text{LIBOR} - 3\%, 0), \quad (4.6)$$

accentuating the fact that the dealer is paying out 3-month LIBOR less the payout from a cap on 3-month LIBOR struck at 3%.⁹ This trade can equivalently be viewed as one in which the dealer is paying 3-month LIBOR and is implicitly purchasing a 5-year cap struck at 3% from the client.

At the time at which the trade is executed, assume that the 5-year swap rate is 1.647% and assume further that the cost of a \$100mm 5-year cap on 3-month LIBOR struck at 3% is \$1,470,000, or equivalently 147 bps of the

⁹Whenever we see a payout formula involving the minimum of two things, it's a pretty good bet that some sort of short option position has been embedded. Most of our option pricing formulas involve determining the value today of a payoff that involves the maximum of two things. The trick is to figure out how to convert a payoff that depends on the minimum of two things into the maximum of two things, as we did in Equation 4.6.

notional.¹⁰ Whereas the dealer would be willing to receive 1.647% in a plain vanilla 5-year swap, the fixed rate in this trade must be changed in order to reflect the fact that the dealer is implicitly buying the cap. Note that the fixed rate must be lowered: the dealer is buying the cap and, instead of paying an upfront amount for it, will be “paying” for it by receiving a fixed rate that is below the 5-year offer side swap rate.

By how much should the fixed rate in this trade be lowered? The fixed rate should be lowered by an amount such that the present value of the off-marketness of the rate is equal to the value of the cap today. To this end, put

$$\begin{aligned} \text{PV} &= 1.47\% \\ n &= 5 \times 2 = 10 \\ i &= 1.647\%/2 = 0.8235\% \end{aligned}$$

and we get $\text{PMT} = 0.154\%$. This payment is a semiannual installment, and we must multiply by 2 in order to annualize. Thus, the appropriate fixed rate in this trade is $1.647\% - (2 \times 0.154\%) = 1.339\%$, as indicated in Figure 4.2.¹¹

4.1.4 Valuing Caps and Floors

In order to value caps and floors, it is common to use an extension of the Black-Scholes formula known as Black’s formula (also called Black’s model). We need to keep a couple of caveats in mind along the way. First, recall that a cap or a floor is a series or portfolio of individual options, called caplets or floorlets, respectively. In order to value caps and floors we value each of the individual options separately, and the value of the cap or floor is just the sum of the values of the individual options. Second, the LIBOR reset for a caplet or floorlet occurs at the beginning of a period, but the actual payment from the option does not occur until the end of the period. This time lag between option expiry and payment is something we may not have seen before. We typically assume that the payment from an in-the-money option occurs at expiry. This payment lag is incorporated in the valuation formula.

To price individual caplets or floorlets, consider a sequence of equally spaced times $t_0, t_1, t_2, \dots, t_n$ and let $\Delta t \equiv t_{i+1} - t_i$ represent the amount of time between LIBOR resets expressed in days. For example, if we are pricing

¹⁰In the market for caps and floors as well as the market for swaptions, it is more common to express the upfront premium of an option in basis points of the notional, rather than a dollar amount. If the dollar premium of the option is \$1,470,000, then the premium as a percentage of the notional is $\$1,470,000/\$100,000,000 = 1.47\% = 147 \text{ bps}$.

¹¹It is important to note that the trade depicted in Figure 4.2 is an NPV = 0 transaction. The fixed rate in the trade has been adjusted from the par swap rate so as to exactly offset the value of the cap that the dealer is implicitly buying.

caplets on 3-month LIBOR, then Δt is the number of days in a 3-month period, and if we are pricing caplets on 1-year LIBOR in our stylized example, then Δt is the number of days in one year, or 360. Define $\tau = \Delta t/360$, the day count fraction. The payoff of a single caplet at time t_{i+1} is equal to

$$\tau \times \text{Notional} \times \max [L(t_i, t_{i+1}) - X, 0], \quad (4.7)$$

where $L(t_i, t_{i+1})$ is the LIBOR that sets at time t_i . Once again, note that the LIBOR rate sets at time t_i but any in-the-money caplet payment is made a period later at time t_{i+1} .

Denote by $f(t_i, t_{i+1})$ the current value of the forward LIBOR rate that sets at time t_i . We will obtain $f(t_i, t_{i+1})$ by bootstrapping today's swap curve. Under the usual Black-Scholes assumptions, including the assumption that the forward LIBOR rate, whose initial value is $f(t_i, t_{i+1})$, is lognormally distributed, we can apply Black's formula for the value of a caplet that has a LIBOR reset at time t_i and pays off at time t_{i+1} :

$$\tau \times DF(0, t_{i+1}) \times [f(t_i, t_{i+1})N(d_1) - XN(d_2)], \quad (4.8)$$

where

$$d_1 = \frac{\ln(f(t_i, t_{i+1})/X) + \sigma_{t_i}^2 \times t_i/2}{\sigma_{t_i} \times \sqrt{t_i}} \quad (4.9)$$

and

$$d_2 = d_1 - \sigma_{t_i} \times \sqrt{t_i}. \quad (4.10)$$

In Equation 4.8, $DF(0, t_{i+1})$ is the discount factor for a payment received at time t_{i+1} —that is, $DF(0, t_{i+1})$ is the number such that the present value of \$1 received at time t_{i+1} is equal to $\$1 \times DF(0, t_{i+1})$; σ_{t_i} is the volatility of the forward LIBOR rate set at time t_i ; and $N(\cdot)$ is the cumulative normal distribution function.¹²

The value of a floorlet is given by

$$\tau \times DF(0, t_{i+1}) \times [XN(-d_2) - f(t_i, t_{i+1})N(-d_1)], \quad (4.11)$$

where d_1 and d_2 are as given in Equations 4.9 and 4.10, respectively. To compute the price of a caplet or floorlet in dollars, we multiply Equation 4.8 or 4.11, respectively, by the notional of the option.

¹²What happened here? We started out by talking about how to value an option on some future unknown LIBOR rate, $L(t_i, t_{i+1})$ in Equation 4.7, and we wind up with an option pricing formula that involves the forward LIBOR rate. The intuition for this is as follows: note that at time t_i , the forward rate and the spot rate must converge to the same thing. That is, at time t_i , $f(t_i, t_{i+1}) = L(t_i, t_{i+1})$. Thus, we can think about transforming the problem from valuing an option on some future LIBOR rate to instead an option on the forward. We know the current value of the forward, $f(t_i, t_{i+1})$, from bootstrapping, assume that the forward itself is lognormally distributed, and apply Black's formula, which is a model that was developed to price options on forwards.

In order to price up the first caplet or floorlet in a spot starting cap or floor, respectively, we would not use Black's formula. In this case, LIBOR has already set, and there is no uncertainty about the payoff at the end of the period. In the case of an in-the-money setting, we simply discount the known payment back to today using the appropriate zero coupon rate.

4.1.5 Vol

The name of the game for dealers that make markets in caps and floors is determining what vol assumptions to use when pricing these options.¹³ One tends to see more disparate prices from dealers quoting markets on caps and floors than in swaps. There are several explanations for this. The first is simply that the vol markets are less liquid: it is harder for dealers to hedge vol trades, resulting in relatively wider bid/ask markets for options. Second, the vol markets tend to be a lot less transparent. The information flow for options pricing tends to be much more limited than for swaps, resulting in disparate pricing among dealers. Third, different dealers may have bona fide differences in the way they model and hedge different option trades. This does not tend to be the case for standard caps and floors but does present itself for even slightly complicated structures.

As we discussed in Section 4.1.4, a dealer who makes markets in caps and floors may choose to make individual vol assumptions for each of the caplets that he prices. The relationship between the time to expiry of an individual caplet and the vol used to price that caplet is known as the *base vol curve*.

Another vol concept pertaining to caps and floors is *average vol*. Using the relevant vols for each caplet or floorlet according to the base vol curve, we obtain an upfront cost for the cap or floor. We could then ask the question: given this upfront cost, if we constrained each of the individual vols used to price up each caplet or floorlet to be the same, what single vol gives us the same upfront cost? For a cap or floor of a given maturity, the answer to this question is *average vol*. The relationship between the final maturity of a cap or floor and its average vol is known as the *average vol curve*.

Figure 4.3 demonstrates what the base vol and average vol curves typically look like. This base vol curve is formed by looking at the implied volatilities of various caplets (or floorlets) of different maturities. Given an option price, the *implied vol* is that level of vol used in an option model such as in Equations 4.8 or 4.11 that equates to the given price. Note that the

¹³In many markets, including the rates markets, the word "volatility" has become abbreviated to just "vol."

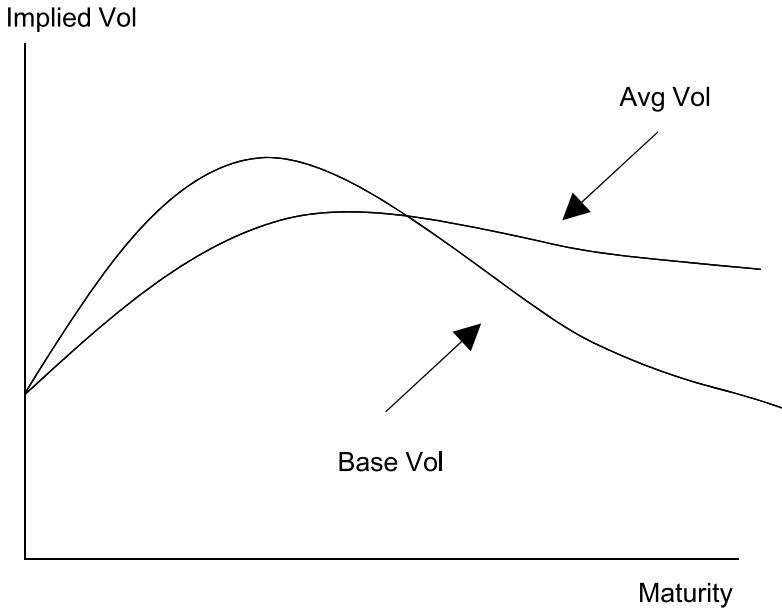


Figure 4.3: Base and Average Vol Curves for Caps and Floors

typical shape of the base vol curve is such that it is upward sloping initially and then tends to decline. That is, caplet volatilities initially tend to increase with an increase in expiration but then eventually decline. The intuition for this “humped” vol curve often suggested by those in industry is as follows: initially, there is some volatility concerning the movement in interest rates. Over longer periods of time, there is more uncertainty in the economy and increased risks of possible shocks to the market, and so on, that add further uncertainty surrounding potential future rates. That is, there is certainly some volatility in rates that we should expect over the next month or two. However, over a longer period of time, there may be the additional uncertainty created, for example, by shifts in Fed policy and uncertain economic conditions that manifests itself in higher implied vols. This would tend to suggest that for longer periods of time, as more uncertainty exists about the future, implied vols would tend to go up. But empirically we usually see that implied vols tend to come down after some point, as depicted in Figure 4.3. The usual explanation for this phenomenon is that interest rates tend to be *mean reverting*, which means that they exhibit the tendency to trend to some long-run mean. Regardless of these intuitive explanations for the shape of the base vol curve for caps and floors, ultimately supply and demand de-

SPOT VOLATILITY			FORWARD VOLATILITY		
Maturity	BID/OFFER	ATM	Maturity	BID/OFFER	ATM
1 year	103.2/104.2	0.49%	1x2	77.7/78.7	0.89%
2 year	85.7/86.7	0.72%	2x3	56.0/57.0	1.55%
3 year	67.7/68.7	1.02%	2x4	47.8/48.8	1.88%
4 year	57.6/58.6	1.33%	3x4	41.3/42.3	2.22%
5 year	49.5/50.5	1.62%	3x5	37.4/38.4	2.49%
7 year	41.2/42.2	2.02%	4x5	35.2/36.2	2.76%
10 year	34.0/35.0	2.51%	5x7	31.0/32.0	3.03%
			5x10	27.7/28.7	3.49%
			7x10	25.6/26.6	3.80%

Table 4.1: Composite Page of Implied Cap/Floor Vols. Used by Permission of Thomson Reuters and BrokerTec

termines the levels at which caps and floors trade and thus at which implied vols trade.¹⁴

Table 4.1 shows a sample composite page of indicative implied vols for caps and floors based on levels from a number of dealers. There are no upfront premia displayed on this page; only implied vols are provided. The first set of numbers labeled “Spot Volatility” is the average vol numbers for caps and floors of various maturities. For example, the market for 5-year caps and floors on 3-month LIBOR is 49.5/50.5. This means that the most aggressive buyer of 5-year caps and floors among dealers will use an average vol of 49.5% to price up the options; all other dealers will use a lower average vol and thus bid the options at lower prices.¹⁵ The most aggressive seller of 5-year caps among dealers will use an average vol of 50.5% to price up caps and floors; all other dealers will use a higher average vol and thus will offer caps and floors at higher prices. Note that each dealer may well be using individual base vols to price up each of the 20 caplets in a 5-year cap. But it would be terribly inconvenient to talk about each of those 20 vols when comparing prices, so it is all rolled up into one number: average vol.

It is important to note that these indicative markets on this composite page only represent markets for caps and floors that are struck at-the-money.

¹⁴Although options in the rates market trade on the basis of price, it has become commonplace to talk about where options trade on the basis of implied vol.

¹⁵Both caps and floors have positive vegas, meaning their cost increases with an increase in volatility.

The at-the-money strike for a cap or a floor is defined to be the mid-market quarterly money par swap rate for the given maturity, which we know from cap-floor parity is the strike for which a cap and floor have the same value. The column labeled “ATM” under the “Spot Volatility” heading stands for *at-the-money* and provides this par swap rate. Thus, we see that the 5-year quarterly money swap rate is 1.62%.¹⁶ For options that have strikes that are not at-the-money, the implied vols obtained from the prices of these options may be different from the implied vols obtained from the prices of at-the-moneys, as we will discuss in Chapter 5.

A *forward starting cap or floor*, by analogy to a forward starting swap, is a cap or floor that is executed today but has its first reset beyond the standard $T + 2$ New York and London Business Days. It is common for dealers to make markets and trade in a variety of forward starting caps and floors as well. If dealers were to have large discrepancies among cap and floor prices, this most likely means that they are not agreeing on the volatility assumptions used to price the trades. Since each dealer is using a whole series of base vols to price the options, these discrepancies may be more prevalent among certain caplets and floorlets than others.

For example, suppose two counterparties are pricing a spot starting 5-year cap. Their respective prices indicate that they are disagreeing on the average vol input. The two counterparties compare their prices for spot starting 3-year caps and find that their prices for these structures are actually very similar. The pricing discrepancy must therefore be in the eight caplets constituting the 2-year cap, three years forward, also known as the 3x5 (pronounced “3 by 5”) cap. The second column of numbers on the composite page labeled “Forward Volatility” gives average vol markets for a variety of forward starting caps and floors. For instance, the average vol market for 3x5 caps is 37.4% on the bid side and 38.4% on the offer side. These vols are applicable only for the at-the-money strike, which is defined to be the quarterly money par swap rate for the corresponding forward starting swap. The column labeled “ATM” under the “Forward Volatility” section provides these at-the-money strikes.

4.1.6 Valuing Caps and Floors in Our Stylized Model

We can apply Black’s formula to price up caps and floors in the context of our stylized model in which we have 1-year LIBOR that resets and pays annually

¹⁶This quarterly money swap rate of 1.62% is equivalent to a quarterly bond swap rate of $1.62\% \times 365/360 = 1.6425\%$, which is equivalent to $2 \times [(1 + 0.016425\%/4)^{4/2} - 1] = 1.646\%$ semi bond. This is higher than the mid-market 5-year semi bond swap rate shown in Table 1.1. The explanation for this is simple: the data from Table 1.1 is from August 11, 2010, whereas the data from Table 4.1 is from August 27, 2010.